

ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

Chapter 5: Instrument Landing System (ILS: LOC, GP & Markers)

Syllabus Source: CAE Oxford (pp. 169-216) | Pass Target: >= 90% AviationExam

1. THE ILS TRI-COMPONENT ARCHITECTURE

The Instrument Landing System provides precision lateral, vertical, and distance guidance during final approach:

ILS Subsystem	Frequency Band & Modulation Carrier	Beam Dimensions & Guaranteed Coverage
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Localizer (LOC)	VHF (108.10 to 111.95 MHz, odd tenths). 90 Hz (Left) / 150 Hz (Right) DDM.	Course width: ~5° (total 350 ft at threshold). Coverage: 35° each side to 17 NM; 10° each side to 25 NM.
Glide Path (GP)	UHF (329.15 to 335.00 MHz, paired with LOC). 90 Hz (Above) / 150 Hz (Below) DDM.	Slope: Standard 3.0°. Beam thickness: ~1.4° (0.7° above/below). Coverage: 8° each side to 10 NM. False slopes at 6° and 9°!
Marker Beacons	VHF 75.0 MHz (vertical fan beam).	Outer (OM: 400 Hz, blue dashes, 4-7 NM); Middle (MM: 1300 Hz, amber dot-dashes, 0.5-0.8 NM); Inner (IM: 3000 Hz, white dots).

2. GLIDE PATH RATE OF DESCENT (ROD) FORMULA

- Master ROD Formula (3.0° Glide Path): $\text{Rate of Descent (ft/min)} = \text{Groundspeed (knots)} \times 5$ (e.g. at 140 kt GS: $\text{ROD} = 140 \times 5 = 700 \text{ ft/min!}$).
- Non-Standard Slope Formula: $\text{ROD (ft/min)} = \text{Groundspeed (knots)} \times \text{Glide Angle (degrees)} \times 100 / 60$.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Target Rate of Descent on ILS

SCENARIO (Everyday Airline Flying Drill): An airliner is established on an ILS approach with a 3.0° glide slope: - Target approach speed (Vref) = 140 knots Indicated Airspeed - Headwind component on final approach = 20 knots QUESTION: What target Rate of Descent (ft/min) must the flight crew maintain on the vertical speed indicator (VSI)? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Groundspeed (GS): - Remember: Rate of descent depends on GROUND SPEED, not airspeed! - Groundspeed = TAS - Headwind - At sea level, TAS ~ IAS = 140 kt. - Groundspeed = 140 kt - 20 kt = 120 knots! Step 2: Recall the Master 3.0° Glide Path Formula: - Rate of Descent (ft/min) = Groundspeed (kt) x 5! Step 3: Calculate Target ROD: - ROD = 120 knots x 5 = 600 ft/min! FINAL ANSWER: Target Rate of Descent is 600 ft/min. If headwind decreases, groundspeed rises and ROD must increase proportionally.

[!] EXAM TRAP (AVIATIONEXAM) - False Glide Path Trap

- False glide paths are generated at HARMONIC MULTIPLES (typically $2 \times 3^\circ = 6^\circ$, and $3 \times 3^\circ = 9^\circ$)! - The first false glide path at 6° has NORMAL SENSING (90 Hz above, 150 Hz below)! - If intercepted from above, you will descend at double the normal rate (1,400 ft/min)!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: At what angle does the lowest FALSE GLIDE PATH occur on a standard 3.0° ILS installation?**

[A] 4.5° [B] At DOUBLE the nominal glide path angle: 6.0 DEGREES [C] 9.0° [D] 1.5°

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. False glide paths occur at multiples of the true angle; the lowest false path is at twice the true slope ($2 \times 3^\circ = 6^\circ$).

Q2: In which frequency band does the ILS GLIDE PATH transmitter operate?

[A] VHF [B] UHF (329.15 to 335.00 MHz) [C] SHF [D] LF

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The Glide Path operates in the UHF band, while the Localizer operates in the VHF band.

Q3: What is the audio modulation frequency and visual color of the ILS OUTER MARKER (OM)?

[A] 1300 Hz amber [B] 400 HZ with BLUE light flashing dashes (two per second) [C] 3000 Hz white [D] Continuous tone red

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The Outer Marker is identified by a 400 Hz tone keyed as low-pitched dashes with a flashing blue light.